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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

4.2 What's Inside a Router?

4.3 Switching

4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

4.5 Introduction to Network Layer Protocols

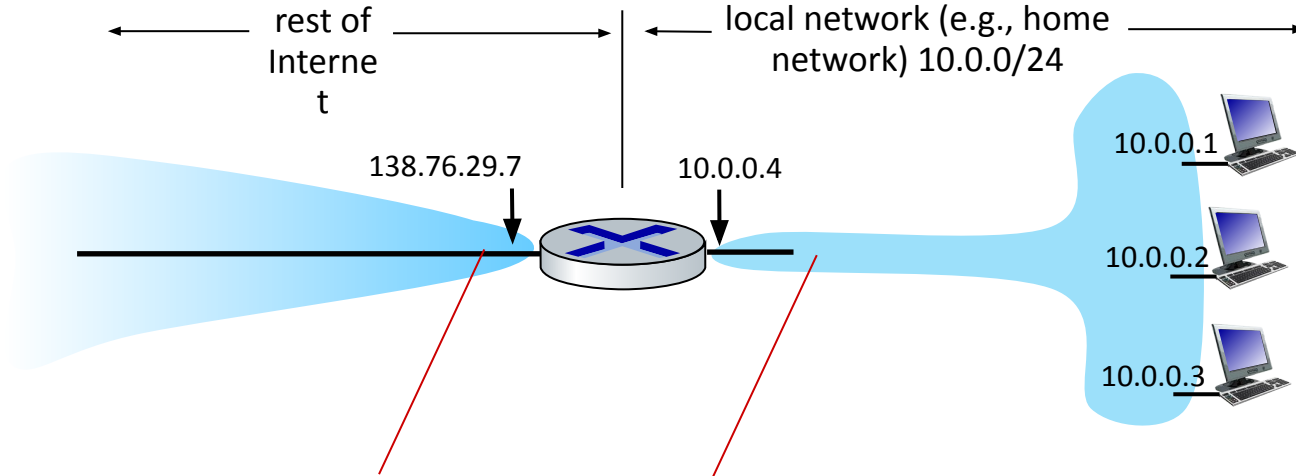
- DHCP
- ICMP

4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

- all devices in local network have 32-bit addresses in a “private” IP address space (10/8, 172.16/12, 192.168/16 prefixes) that can only be used in local network
- advantages:
 - just **one** IP address needed from provider ISP for *all* devices
 - can change addresses of host in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - security: devices inside local net not directly addressable, visible by outside world

NAT: all devices in local network share just **one** IPv4 address as far as outside world is concerned



all datagrams *leaving* local network have *same* source NAT IP address: 138.76.29.7, but *different* source port numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

implementation: NAT router must (transparently):

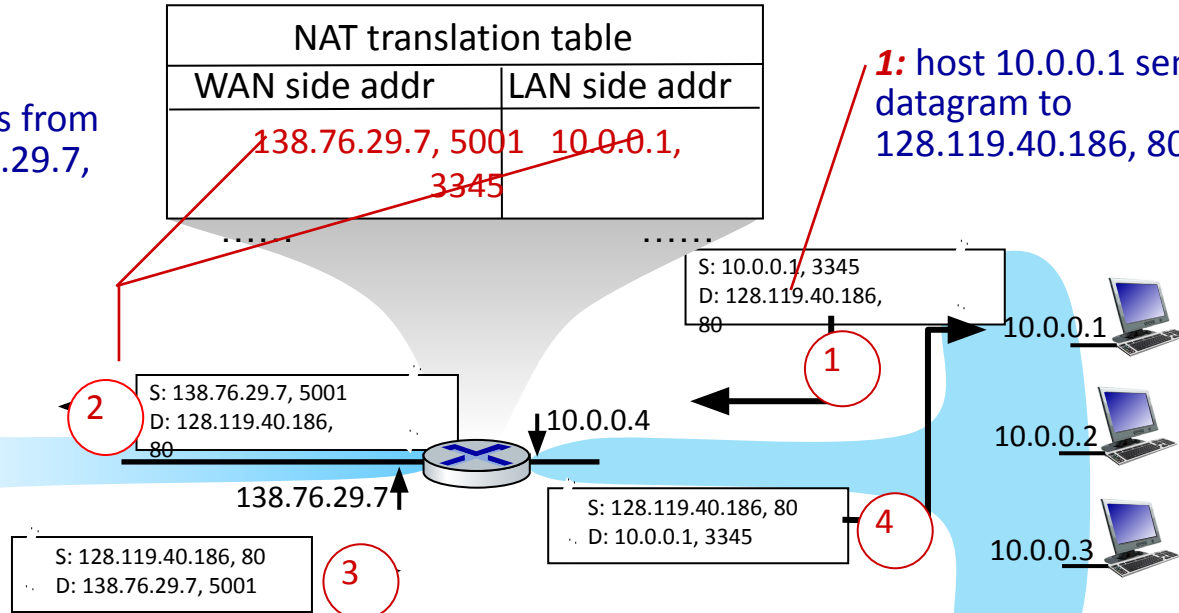
- **outgoing datagrams: replace** (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - remote clients/servers will respond using (NAT IP address, new port #) as destination address
- **remember (in NAT translation table)** every (source IP address, port #) to (NAT IP address, new port #) translation pair
- **incoming datagrams: replace** (NAT IP address, new port #) in destination fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

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NAT: Network Address Translation

2: NAT router changes datagram source address from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table

1: host 10.0.0.1 sends datagram to 128.119.40.186, 80

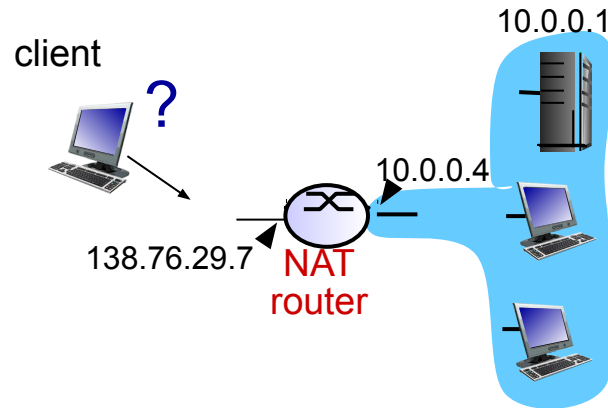


3: reply arrives, destination address: 138.76.29.7, 5001

- NAT has been controversial:
 - routers “should” only process up to layer 3
 - address “shortage” should be solved by IPv6
 - violates end-to-end argument (port # manipulation by network-layer device)
 - NAT traversal: what if client wants to connect to server behind NAT?
- but NAT is here to stay:
 - extensively used in home and institutional nets, 4G/5G cellular nets

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NAT: Network Address Translation



Solution: statically configure NAT
to forward incoming connection
requests at given port to server
e.g., (123.76.29.7, port 25000)
always forwarded to 10.0.0.1 port
25000



THANK YOU

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